

```
In [5]: import pandas as pd
import numpy as np
```

```
In [6]: df = pd.read_csv("train (3).csv")
df.head()
```

Out[6]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500

```
In [7]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
In [10]: df.shape
```

```
Out[10]: (891, 12)
```

```
In [11]: df.describe
```

```
Out[11]: <bound method NDFrame.describe of
0      1      0      3
1      2      1      1
2      3      1      3
3      4      1      1
4      5      0      3
..      ...      ...      ...
886     887      0      2
887     888      1      1
888     889      0      3
889     890      1      1
890     891      0      3

                                Name      Sex  Age  SibSp  \
0                                Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                                Heikkinen, Miss. Laina  female  26.0      0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                                Allen, Mr. William Henry    male  35.0      0
..      ...      ...      ...      ...
886                                Montvila, Rev. Juozas    male  27.0      0
887                                Graham, Miss. Margaret Edith  female  19.0      0
888  Johnston, Miss. Catherine Helen "Carrie"  female   NaN      1
889                                Behr, Mr. Karl Howell    male  26.0      0
890                                Dooley, Mr. Patrick    male  32.0      0

      Parch      Ticket      Fare Cabin Embarked
0         0      A/5 21171   7.2500   NaN        S
1         0      PC 17599  71.2833   C85        C
2         0  STON/O2. 3101282   7.9250   NaN        S
3         0      113803  53.1000  C123        S
4         0      373450   8.0500   NaN        S
..      ...      ...      ...      ...      ...
886         0      211536  13.0000   NaN        S
887         0      112053  30.0000  B42        S
888         2  W./C. 6607   23.4500   NaN        S
889         0      111369  30.0000  C148        C
890         0      370376   7.7500   NaN        Q
```

[891 rows x 12 columns]>

```
In [14]: #mean
df.mean(numeric_only=True)
```

```
Out[14]: PassengerId    446.000000
Survived      0.383838
Pclass        2.308642
Age           29.699118
SibSp         0.523008
Parch         0.381594
Fare          32.204208
dtype: float64
```

```
In [15]: #Median
df.median(numeric_only=True)
```

```
Out[15]: PassengerId    446.0000  
         Survived      0.0000  
         Pclass        3.0000  
         Age          28.0000  
         SibSp         0.0000  
         Parch         0.0000  
         Fare         14.4542  
         dtype: float64
```

```
In [17]: #Mode  
df.mode()
```

Out[17]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	C
0	1	0.0	3.0	Abbing, Mr. Anthony	male	24.0	0.0	0.0	1601	8.05	
1	2	NaN	NaN	Abbott, Mr. Rossmore Edward	NaN	NaN	NaN	NaN	347082	NaN	
2	3	NaN	NaN	Abbott, Mrs. Stanton (Rosa Hunt)	NaN	NaN	NaN	NaN	CA. 2343	NaN	
3	4	NaN	NaN	Abelson, Mr. Samuel	NaN	NaN	NaN	NaN	NaN	NaN	
4	5	NaN	NaN	Abelson, Mrs. Samuel (Hannah Wizosky)	NaN	NaN	NaN	NaN	NaN	NaN	
...	...	...	...	...	...	...	...	...	...	...	...
886	887	NaN	NaN	de Mulder, Mr. Theodore	NaN	NaN	NaN	NaN	NaN	NaN	
887	888	NaN	NaN	de Pelsmaecker, Mr. Alfons	NaN	NaN	NaN	NaN	NaN	NaN	
888	889	NaN	NaN	del Carlo, Mr. Sebastiano	NaN	NaN	NaN	NaN	NaN	NaN	
889	890	NaN	NaN	van Billiard, Mr. Austin Blyler	NaN	NaN	NaN	NaN	NaN	NaN	
890	891	NaN	NaN	van Melkebeke, Mr. Philemon	NaN	NaN	NaN	NaN	NaN	NaN	

891 rows × 12 columns



In [18]:

```
df.dtypes
```

```
Out[18]: PassengerId      int64
Survived      int64
Pclass        int64
Name          object
Sex           object
Age           float64
SibSp         int64
Parch         int64
Ticket        object
Fare          float64
Cabin         object
Embarked      object
dtype: object
```

```
In [19]: df.groupby('Sex')['Age'].mean()
```

```
Out[19]: Sex
female    27.915709
male      30.726645
Name: Age, dtype: float64
```

```
In [20]: df.groupby('Pclass')['Fare'].mean()
```

```
Out[20]: Pclass
1      84.154687
2      20.662183
3      13.675550
Name: Fare, dtype: float64
```

```
In [21]: df.isnull().sum()
```

```
Out[21]: PassengerId      0
Survived      0
Pclass        0
Name          0
Sex           0
Age           177
SibSp         0
Parch         0
Ticket        0
Fare          0
Cabin         687
Embarked      2
dtype: int64
```

```
In [24]: df['Age'] = df['Age'].fillna(df['Age'].mean())
```

```
In [25]: df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
```

```
In [26]: df.isnull().sum()
```

```
Out[26]: PassengerId      0
         Survived        0
         Pclass          0
         Name            0
         Sex             0
         Age             0
         SibSp           0
         Parch           0
         Ticket          0
         Fare            0
         Cabin          687
         Embarked        0
         dtype: int64
```

```
In [29]: #Convert Categorical Data to Numeric
         from sklearn.preprocessing import LabelEncoder

         le = LabelEncoder()
         df['Sex'] = le.fit_transform(df['Sex'])
```

```
In [31]: df.columns
```

```
Out[31]: Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
               'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked_Q', 'Embarked_S'],
              dtype='object')
```

```
In [35]: if 'Embarked' in df.columns:
         df = pd.get_dummies(df, columns=['Embarked'], drop_first=True)
```

```
In [36]: from sklearn.preprocessing import OneHotEncoder

         enc = OneHotEncoder()
         enc_df = pd.DataFrame(enc.fit_transform(df[['Pclass']]).toarray())
         enc_df.head()
```

```
Out[36]:
```

	0	1	2
0	0.0	0.0	1.0
1	1.0	0.0	0.0
2	0.0	0.0	1.0
3	1.0	0.0	0.0
4	0.0	0.0	1.0

```
In [37]: df_encoded = df.join(enc_df)
         df_encoded.head()
```

Out[37]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	1	22.0	1	0	A/5 21171	7.2500
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	0	38.0	1	0	PC 17599	71.2833
2	3	1	3	Heikkinen, Miss. Laina	0	26.0	0	0	STON/O2. 3101282	7.9250
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	0	35.0	1	0	113803	53.1000
4	5	0	3	Allen, Mr. William Henry	1	35.0	0	0	373450	8.0500

In [38]:

```
df.min(numeric_only=True)
df.max(numeric_only=True)
```

Out[38]:

PassengerId

891

Survived

1

Pclass

3

Sex

1

Age

80.0

SibSp

8

Parch

6

Fare

512.3292

Embarked_Q

True

Embarked_S

True

dtype: object

In []: